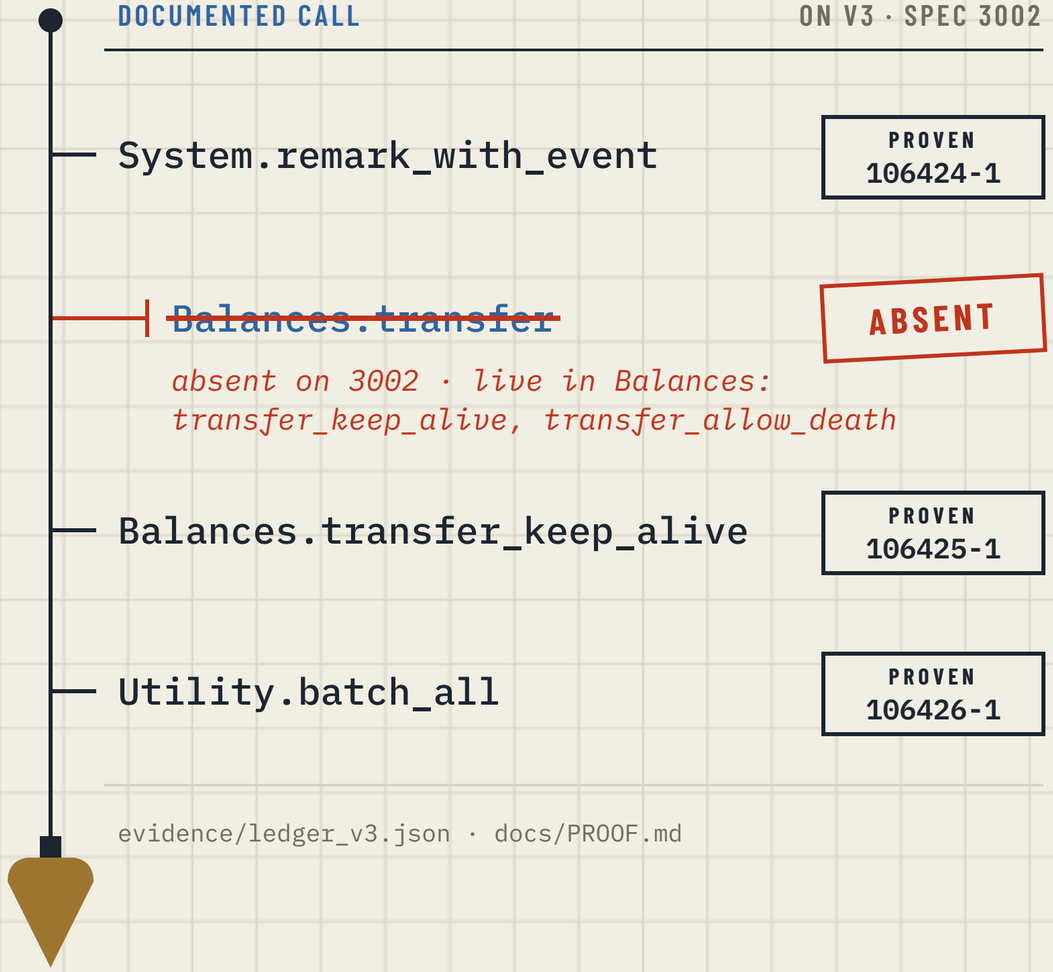
Plumbline

Portaldot's documented calls, plumbed against the V3 runtime and proven by real transactions.

One verdict per documented entry on the V3 testnet: 549 verdict rows, and 3 canary transactions read back from their blocks.

BUILT FOR PORTALDOT HACKER HOUSE 2026	MODULAR CLOUD & MIDDLEWARE	v0.0.1



DOCS AND CHAIN DISAGREE

A builder at the Hacker House copies `Balances.transfer` from the Portaldot docs. On the V3 testnet that call does not exist, and the page does not say which runtime it describes.

Today the way to find out is to submit a transaction and read the error. The next runtime upgrade moves the line again.

	WHAT THE SOURCES SAID	WHAT V3 REPORTS
ENDPOINT	~~testnetv3-node.feso-apps.xyz~~ V3 testnet guide §1.1 · no DNS on 2026-10-14	test-node-v3.portaldot.dev
SPEC	~~portaldotV3 / 3000~~ a third-party README, not re-verified	portaldot-v3 / 3002
SYMBOL	~~tPOTv3~~ V3 testnet guide §1.1, §3.1	tPOT
PALLETS	~~70~~ the same README, not re-verified	69

79 of the 196 calls in Portaldot's mainnet docs disagree with mainnet spec 1002 (64 absent or renamed, 15 with other arguments).

2 of the same 196 disagree with V3, spec 3002. The pages name no spec, so only plumbing tells which runtime they fit.

CALL, VERDICT, PROOF

STA. 01

DOCUMENTED CALL

22 mainnet extrinsic pages give 196 calls. The V3 testnet guide at **d671a32** gives 46 **compose_call** sites and 12 FAQ modules, read by AST and never executed.

```
$ plumbline docs
$ plumbline guide
```

→

STA. 02

VERDICT

Each entry is joined to one runtime's metadata and gets exactly one verdict, stamped with profile, spec version and metadata hash. Same inputs, byte-identical ledger.

```
$ plumbline plumb
```

→

STA. 03

PROOF

A canary extrinsic is signed and sent, its receipt kept, then re-read from its block by hash in a new process. PROVEN needs that receipt.

```
$ plumbline canary
$ plumbline readback
```

PROVEN | UNPROVEN | ARGS-UNCHECKED | ARG_DRIFT | DRIFTED | ABSENT

verdicts from `plumbline/verdicts.py`

Beside the flow: **plumbline diff A B** lists what moved between two runtimes; **plumbline check <dApp> --target v3** classifies a dApp's call sites against one, with exit codes 0 / 1 / 2.

THE V3 SHEET, REDLINED

Mainnet docs and the V3 guide's FAQ, plumbed against **portaldot-v3** / spec 3002 on 2026-10-19.

DOCUMENTED	VERDICT ON V3
~~Balances.transfer~~ *absent on 3002 · live: transfer_keep_alive, transfer_allow_death*	ABSENT
Balances.transfer_keep_alive canary sent to itself, read back from block 106425: MATCH	PROVEN 106425-1
ElectionProviderMultiPhase.governance_fallback ~~(maybe_max_voters, maybe_max_targets)~~ *takes no arguments on 3002*	ARG_DRIFT
~~conviction_voting · cumulus_parachain_system · ibridge · paras · referenda~~ *FAQ lists them for V3.0 · no pallet by that name*	ABSENT ×5
System.remark_with_event	PROVEN 106424-1
Utility.batch_all	PROVEN 106426-1

MAINNET DOCS ON V3

PROVEN	3
UNPROVEN present, no tx yet	191
ARG_DRIFT	1
ABSENT	1
documented calls	196

> **194 of the 196 hold on V3 by name and arguments; 191 of them wait for a canary.**

The 5 FAQ modules are a separate source (the V3 guide), outside the 196. All sources: 549 verdict rows, each stamped profile `v3` · spec 3002 · metadata hash; `evidence/ledger_v3.json`, blake2-256 `dc808404…1e34`, byte-identical on re-run.

THREE TRANSACTIONS, READ BACK

Portaldot V3 testnet, **portaldot-v3** / spec 3002 · signer **5EhP3KWt…RUzDsb** · canary run **20261019T182014Z**

CALL	EXTRINSIC	TX HASH	NODE-CONSOLE
✓ **System.remark_with_event**	**106424-1**	0xf54430571069e41bfe629c6c77609a279d0ad82fa2a83e45b0c3575bc941e6f5	**#106424 ↗**
✓ **Balances.transfer_keep_alive**	**106425-1**	0xa7effe5d5ab7d7c512690d285065fd83919f8b83dc04de1630e1e4ddff91d90b	**#106425 ↗**
✓ **Utility.batch_all**	**106426-1**	0x25e0352284f3a2455e03c7c02c625b8bd1ff692e244bafbc51d1adf20f5dfaa8	**#106426 ↗**

evidence/readback_v3.txt

```
# plumbline readback evidence/canary_v3.json — profile v3, run 2026-10-19T18:24:33Z, new process, new connection
106424-1 System.remark_with_event      decoded System.remark_with_event      block 106424  success True  MATCH
106425-1 Balances.transfer_keep_alive  decoded Balances.transfer_keep_alive  block 106425  success True  MATCH
106426-1 Utility.batch_all             decoded Utility.batch_all             block 106426  success True  MATCH
profile v3 — Portaldot V3 testnet: 3/3 receipts match the chain
```

The block is fetched by hash; the extrinsic at the receipt's index must carry the tx hash and decode to the same call. node-console looks blocks up by number or hash (no tx-hash search). A local stand-in run is kept as test input and labelled *interim-sdk, not V3*. Receipts: **evidence/canary_v3.json**, **docs/PROOF.md**.

SOURCES, ENGINE, EVIDENCE

in the repo · planned · stand-in, not V3

SOURCES

PORTALDOT RUNTIMES

V3 testnet · spec 3002
read-only metadata + 3 canary txs

mainnet · spec 1002
read-only baseline, never a tx

interim-sdk
local stand-in, not V3, test input

DOCUMENTED SURFACE

portaldot-dev.readthedocs.io
22 extrinsic pages, fetched at run time

V3 guide @ d671a32
cloned outside the repo, AST only

a dApp directory
.py .ts .js call sites + wasm imports

ENGINE · plumbline/

metadata.py
chain profiles · identity check · pinned-block read

canary.py
sign · submit · keep the receipt · re-read by block hash

receipts: PROVEN needs one

plumb.py
SQLite drift ledger · one verdict per entry · canonical export

documented calls

docs.py · guide.py
three page layouts · README table · AST call sites · FAQ

diff.py
two runtimes · normalise V13 / V14+ · six diff kinds

check.py
a dApp against a runtime · exit codes 0 / 1 / 2

profile from metadata.py

PLANNED · Plumbline Ledger contract (Revive) on V3

EVIDENCE, COMMITTED

fixtures/metadata_v3.scale
+ metadata_1002.scale, sha256 recorded

evidence/canary_v3.json
+ docs/PROOF.md · 3 txs, read back 3/3

evidence/ledger_v3.json
549 verdict rows + blake2-256

evidence/guide_calls.json
+ docs_1002.json · names + file:line only

evidence/diff_1002_v3.json
516 rows, mainnet 1002 → V3

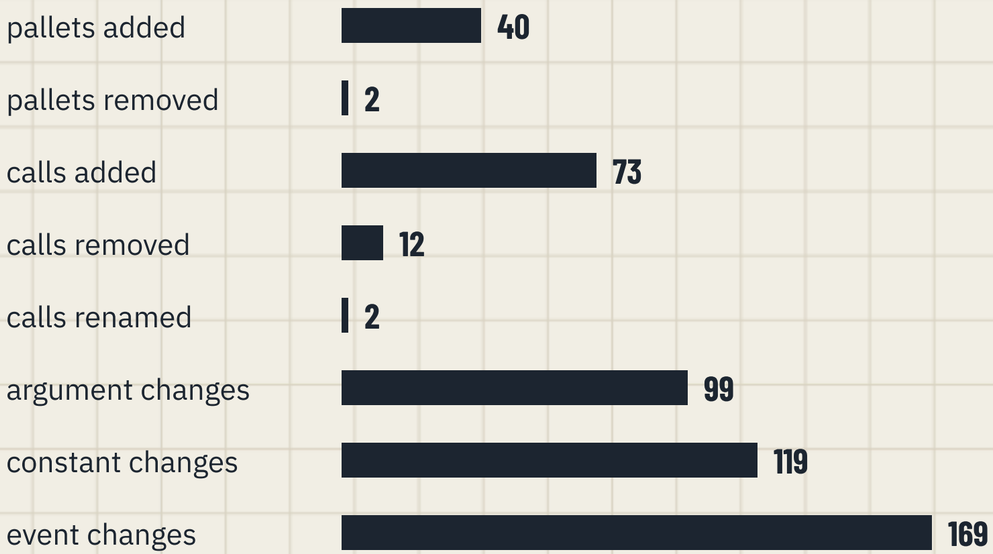
fixtures/expected_check_v3.txt
golden check output

A VERDICT NEEDS A RECEIPT

WHAT EACH SOURCE TELLS YOU ALONE

DOCS PAGES	readthedocs and the V3 guide: what was designed; the pages name no runtime spec
NODE-CONSOLE	Portaldot's explorer: what exists at one block, not what the docs promised
A METADATA DIFF	what moved between two runtimes, not whether the docs match either one
ONE TEST TX	that one call worked once, on one runtime
PLUMBLINE	**all three joined per documented call: one verdict, stamped with the spec, PROVEN only with a receipt, re-run on the next runtime**

WHAT MOVED, MAINNET SPEC 1002 → V3 SPEC 3002



516 rows from `plumbline diff` · `evidence/diff_1002_v3.json`

5 → 0 blocking findings for the same example dApp under `plumbline check`: on spec 1002, then on V3

BUILT ON PORTALDOT'S OWN SURFACE

PORTALDOT'S DOCUMENTED SURFACE

portaldot-dev.readthedocs.io extrinsic pages · the V3 testnet guide and its FAQ

PLUMBLINE · THE MIDDLEWARE BETWEEN THEM

plumb · **diff** · **check** · **canary**

THE SDK PORTALDOT'S DOCS POINT TO

substrate-interface 1.8.1: **compose_call** · **create_signed_extrinsic** · **submit_extrinsic** · **get_block** · **get_events**

PORTALDOT'S TWO RUNTIMES

V3 testnet **portaldot-v3**/3002 · mainnet spec 1002, read-only: **state_getMetadata** · **state_getRuntimeVersion** · **state_getStorageHash** (**:code**)

TRACK: MODULAR CLOUD & MIDDLEWARE

Plumbline sits between Portaldot's docs and its runtime; **check** runs a dApp's calls against V3 before anything is sent.

PINNED TO THE REAL V3

The **v3** profile checks spec name **portaldot-v3**, genesis **0xd4f9acbc…5283bd** and spec 3002, and fails loudly on a mismatch; the **:code** hash is recorded in **evidence/metadata_v3.json**.

RE-RUN PER UPGRADE

The spec 1002 baseline and the spec 3002 sheet come from the same commands; the next runtime is the next run.

WHO RUNS IT, WHO PAYS

WHO	MOMENT OF PAIN	WHAT THEY GET	STATUS
HACKER HOUSE BUILDERS NOW, THE FIRST AUDIENCE	a documented call fails on V3	`plumbline check <dApp> --target v3`; exit code 1 on blocking findings, so it can gate CI	TODAY
TEAMS WITH MAINNET CODE MOVING 1002 → V3	not knowing what breaks on V3	the 516-row runtime diff, and `check` run on their repo	TODAY
PORTALDOT DOCS & DEVREL CUSTOMER #1	re-verifying every page per runtime by hand	the ledger as docs CI: a "verified on spec N" line per reference entry	PLANNED
WALLETS & INTEGRATORS AFTER THE EVENT	an upgrade changes call shapes	a diff and a canary re-run per upgrade; always-on alerts as the paid tier	PLANNED

MODEL

Docs CI as an ecosystem-funded public good, plus paid upgrade alerts for integrators. Portaldot is the first chain; any Substrate chain whose docs trail its runtime has the same gap.

HONEST LIMIT

No revenue and no external users yet. The first two rows work today on the public repo; the last two are the plan.

RE-RUN EVERY NUMBER HERE

AT THE CHECK-IN

- **runtime**, **plumb**, **diff**, **check**, **canary** run against V3, spec 3002
- 549 verdict rows, byte-identical on re-run
- 3 V3 transactions, read back 3/3
- 263 tests passed (2026-10-20) · 86 commits from 2026-10-14 to build 7fd4192

NEXT, BEFORE THE FINAL (NOV 6)

- planned: a canary per documented call (3 of 196 PROVEN today)
- planned: Plumbline Ledger contract (Revive) on V3
- open gap: **check** does not detect TYPE-TRAP yet
- open gap: runtime APIs (ReviveApi) are not checked yet

```
$ pip install -e '.[dev]' && pytest -q
$ plumbline check examples/mainnet-dapp --target v3   # blocking 0
```

AI assistance: Claude Code drafted most of the code (the author reviews it; see the README's Team section) and drafted this deck from the repo's evidence.

Scan: the public repository

REPOSITORY
github.com/edycutjong/portaldot-hackathon-2026-plumbline-edycu

ON-CHAIN PROOF · V3 FINDINGS
docs/PROOF.md · docs/BASELINE-V3.md

CHECK-IN VIDEO
youtu.be/nBi7jbwwi-k